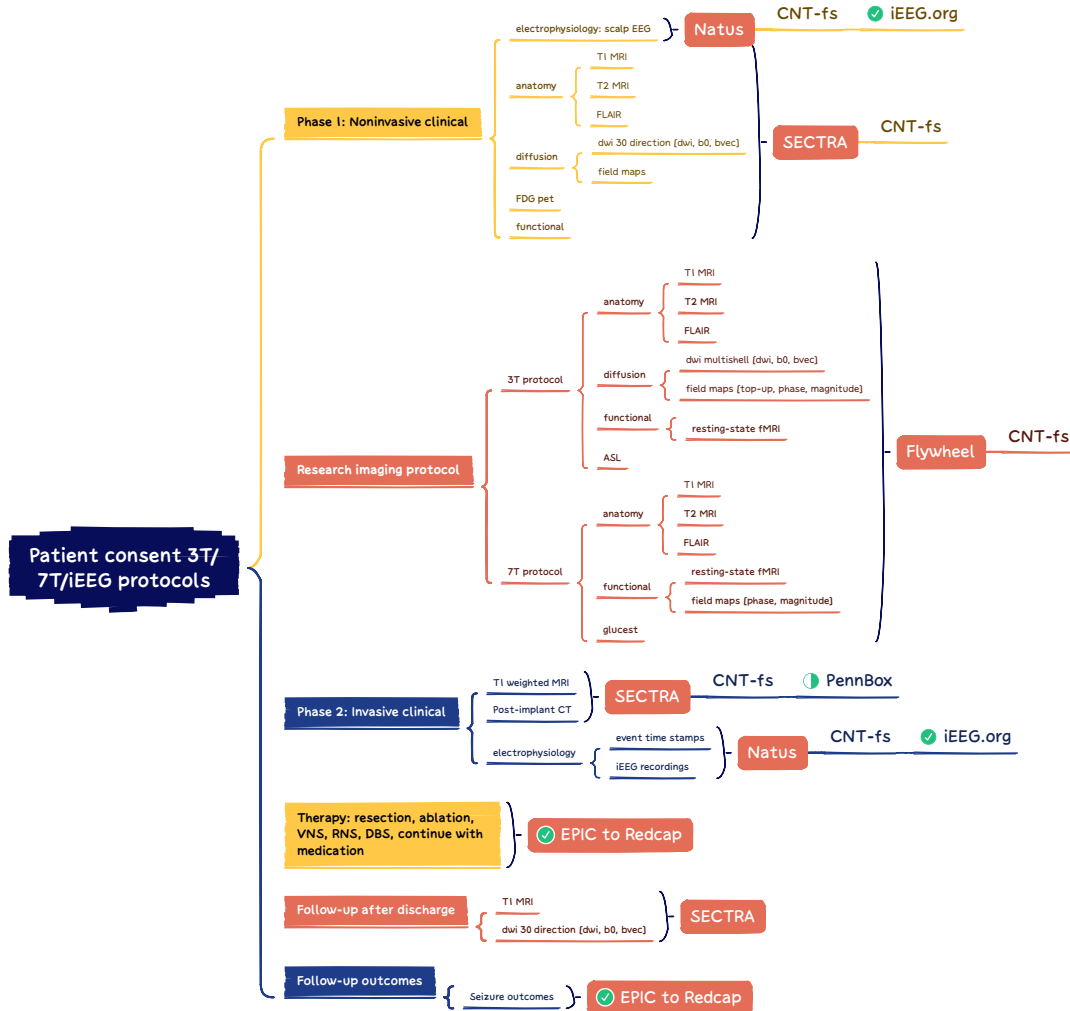


## **CNT data organization**

Nishant Sinha

# Data collection stages and storage



# Making data accessible for research

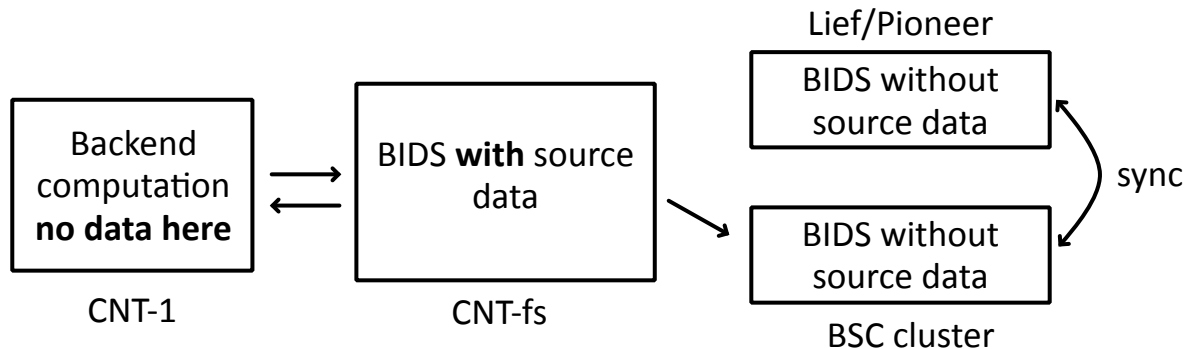
- A researcher in CNT will look for data in:
  - **Leif or Pioneer:** /data/Human\_Data/CNT\_iEEG\_BIDS
  - **BSC cluster:** /project/davis\_group/DATA/BIDS/PIER
- The data in these folders should be in BIDS structure
  - Scanner → dicom files → nifti conversion → de-identification → BIDS
- De-identification: scrub patient identifiers from json files.

| Folders                                                                                  | Folders                                                                                         | Folders                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Folders                                                                                                                                                                                                                                                                                                                                                                     | Other                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  BIDS > |  sub-RID0652 > |  derivatives ><br> ses-clinicalP1 ><br> ses-clinicalP2 ><br> ses-followup1 ><br> ses-research3T ><br> ses-research7T > |  anat ><br> dwi ><br> fmap ><br> func > |  sub-RID...AIR.json<br> sub-RID...IR.nii.gz<br> sub-RID...T1w.json<br> sub-RID...1w.nii.gz<br> sub-RID...T2w.json<br> sub-RID...2w.nii.gz |

*Figure: Desired data organisation*

# Problem

- Typically the data organisation so far has been primarily done by grad students and postdocs.
- Data organisation is time consuming, so students and postdocs have focused primarily on organising the subset of data we need.
- This leaves lots of data unorganised and in some cases data loss!
- To prevent this we need help by research coordinators.
  - Data already collected: complete missing data
  - Data being collected: harmonise with the existing data structure and naming conventions.



## What we have already done?

BSC cluster: /project/davis\_group/DATA/BIDS/PIER

```

[nishants@bcsclub1 BIOS]$ ls
CLINICAL_derivatives flywheel HCP imagingProtocols MIER PIER README sourcedata
[nishants@bcsclub1 BIOS]$ ls PIER/
dataset_description.json sub-RID0859 sub-RID09142 sub-RID09222 sub-RID09287 sub-RID09338 sub-RID09394 sub-RID09455 sub-RID09526 sub-RID09583 sub-RID09619 sub-RID09640 sub-RID09653 sub-RID09854
sub-RID09859 sub-RID09913 sub-RID10013 sub-RID10043 sub-RID10088 sub-RID10132 sub-RID10139 sub-RID10149 sub-RID10156 sub-RID10169 sub-RID10184 sub-RID10185 sub-RID10186 sub-RID10187
sub-RID0913 sub-RID09854 sub-RID10044 sub-RID10236 sub-RID10289 sub-RID10334 sub-RID10405 sub-RID10462 sub-RID10529 sub-RID10587 sub-RID10629 sub-RID10662 sub-RID10688 sub-RID10840
sub-RID0915 sub-RID09855 sub-RID10153 sub-RID10238 sub-RID10290 sub-RID10337 sub-RID10408 sub-RID10472 sub-RID10530 sub-RID10588 sub-RID10633 sub-RID10663 sub-RID10689 sub-RID10884
sub-RID10818 sub-RID10958 sub-RID10154 sub-RID10248 sub-RID10291 sub-RID10338 sub-RID10412 sub-RID10475 sub-RID10534 sub-RID10589 sub-RID10636 sub-RID10664 sub-RID10693 sub-RID10887
sub-RID0920 sub-RID09859 sub-RID10157 sub-RID10241 sub-RID10292 sub-RID10331 sub-RID10415 sub-RID10476 sub-RID10536 sub-RID10591 sub-RID10638 sub-RID10665 sub-RID10697 sub-RID10894
sub-RID0921 sub-RID10065 sub-RID10160 sub-RID10249 sub-RID10294 sub-RID10352 sub-RID10416 sub-RID10477 sub-RID10547 sub-RID10592 sub-RID10639 sub-RID10666 sub-RID10786 sub-RID10911
sub-RID0924 sub-RID10068 sub-RID10165 sub-RID10258 sub-RID10295 sub-RID10356 sub-RID10420 sub-RID10498 sub-RID10595 sub-RID10642 sub-RID10667 sub-RID10815 sub-RID10912
sub-RID0925 sub-RID10069 sub-RID10171 sub-RID10252 sub-RID10296 sub-RID10357 sub-RID10424 sub-RID10499 sub-RID10596 sub-RID10643 sub-RID10668 sub-RID10816 sub-RID10913
sub-RID0929 sub-RID10089 sub-RID10175 sub-RID10259 sub-RID10297 sub-RID10365 sub-RID10429 sub-RID10495 sub-RID10562 sub-RID10597 sub-RID10646 sub-RID10669 sub-RID10817 sub-RID10919
sub-RID0930 sub-RID10101 sub-RID10179 sub-RID10267 sub-RID10307 sub-RID10371 sub-RID10433 sub-RID10502 sub-RID10564 sub-RID10598 sub-RID10647 sub-RID10678 sub-RID10818 sub-RID10927
sub-RID0931 sub-RID10102 sub-RID10186 sub-RID10278 sub-RID10309 sub-RID10372 sub-RID10440 sub-RID10505 sub-RID10565 sub-RID10599 sub-RID10648 sub-RID10671 sub-RID10819 sub-RID109XX
sub-RID0932 sub-RID10106 sub-RID10190 sub-RID10272 sub-RID10313 sub-RID10380 sub-RID10442 sub-RID10508 sub-RID10566 sub-RID10600 sub-RID10649 sub-RID10672 sub-RID10820
sub-RID0933 sub-RID10117 sub-RID10192 sub-RID10274 sub-RID10317 sub-RID10381 sub-RID10445 sub-RID10512 sub-RID10571 sub-RID10602 sub-RID10650 sub-RID10673 sub-RID10825
sub-RID0937 sub-RID10119 sub-RID10193 sub-RID10278 sub-RID10320 sub-RID10382 sub-RID10448 sub-RID10513 sub-RID10572 sub-RID10603 sub-RID10651 sub-RID10675 sub-RID10826
sub-RID0942 sub-RID10121 sub-RID10194 sub-RID10279 sub-RID10322 sub-RID10383 sub-RID10449 sub-RID10517 sub-RID10574 sub-RID10604 sub-RID10652 sub-RID10679 sub-RID10827
sub-RID0946 sub-RID10123 sub-RID10194 sub-RID10280 sub-RID10325 sub-RID10385 sub-RID10450 sub-RID10518 sub-RID10576 sub-RID10605 sub-RID10658 sub-RID10670 sub-RID10830
sub-RID0948 sub-RID10131 sub-RID10213 sub-RID10285 sub-RID10328 sub-RID10386 sub-RID10452 sub-RID10528 sub-RID10578 sub-RID10614 sub-RID10658 sub-RID10681 sub-RID10833
sub-RID0949 sub-RID10139 sub-RID10218 sub-RID10286 sub-RID10329 sub-RID10392 sub-RID10454 sub-RID10522 sub-RID10582 sub-RID10615 sub-RID10659 sub-RID10682 sub-RID10839

[nishants@bcsclub1 BIOS]$ cd sub-RID0815
-bash: cd: sub-RID0815: No such file or directory
[nishants@bcsclub1 BIOS]$ cd PIER/sub-RID0815
[nishants@bcsclub1 sub-RID0815]$ ls
research3Tv04
[nishants@bcsclub1 sub-RID0815]$ cd ses
-bash: cd: ses: No such file or directory
[nishants@bcsclub1 sub-RID0815]$ cd ses-research3Tv04/
[nishants@bcsclub1 ses-research3Tv04]$ ls
anat dwi fmap func
[nishants@bcsclub1 ses-research3Tv04]$ cd anat/
[nishants@bcsclub1 anat]$ ls
sub-RID0815_ses-research3Tv04_acq-3D_FLAIR.json sub-RID0815_ses-research3Tv04_acq-3D_T1w.json sub-RID0815_ses-research3Tv04_acq-3D_T2w.json
sub-RID0815_ses-research3Tv04_acq-3D_FLAIR.nii.gz sub-RID0815_ses-research3Tv04_acq-3D_T1w.nii.gz sub-RID0815_ses-research3Tv04_acq-3D_T2w.nii.gz
[nishants@bcsclub1 anat]$

```

Leif or Pioneer: /data/Human Data/CNT iEEG BIDS

| nishants@pioneer:/data/Human_Data/ cnt_1s |             |               |             |                        |             |             |             |             |                             |                 |             |             |             |             |
|-------------------------------------------|-------------|---------------|-------------|------------------------|-------------|-------------|-------------|-------------|-----------------------------|-----------------|-------------|-------------|-------------|-------------|
| archive_TNL_Data.tar.gz                   | BIDS        | CNT_iEEG_BIDS | CounterAct  | HumanNV                | HyperTing   | Isa_Pioneer | NeuroPace   | PREVENT     | S_tressorTig                | VirtualReaction |             |             |             |             |
| archive_UCSF_I008_P001.tar.gz             | BIDS_0ld    | CHOP          | Columbia    | Electrode_Localization | HUP         | Kahane      | Mayo        | PET_niftis  | Study019_T1_20000425.nii.gz | TACERN_Sample   |             |             |             |             |
| nishants@pioneer:/data/Human_Data/ cnt_1s |             |               |             |                        |             |             |             |             |                             |                 |             |             |             |             |
| nishants@pioneer:/data/Human_Data/ cnt_1s |             |               |             |                        |             |             |             |             |                             |                 |             |             |             |             |
| nishants@pioneer:/data/Human_Data/ cnt_1s |             |               |             |                        |             |             |             |             |                             |                 |             |             |             |             |
| sub-RID0013                               | sub-RID0042 | sub-RID0070   | sub-RID0166 | sub-RID0213            | sub-RID0279 | sub-RID0295 | sub-RID0330 | sub-RID0392 | sub-RID0476                 | sub-RID0566     | sub-RID0615 | sub-RID0679 | sub-RID0820 | sub-RID0911 |
| sub-RID0014                               | sub-RID0046 | sub-RID0089   | sub-RID0165 | sub-RID0230            | sub-RID0280 | sub-RID0296 | sub-RID0332 | sub-RID0394 | sub-RID0490                 | sub-RID0572     | sub-RID0617 | sub-RID0682 | sub-RID0825 | sub-RID0912 |
| sub-RID0015                               | sub-RID0050 | sub-RID0181   | sub-RID0167 | sub-RID0238            | sub-RID0282 | sub-RID0297 | sub-RID0334 | sub-RID0405 | sub-RID0502                 | sub-RID0582     | sub-RID0621 | sub-RID0683 | sub-RID0826 | sub-RID0913 |
| sub-RID0018                               | sub-RID0051 | sub-RID0182   | sub-RID0171 | sub-RID0240            | sub-RID0285 | sub-RID0301 | sub-RID0337 | sub-RID0412 | sub-RID0505                 | sub-RID0583     | sub-RID0646 | sub-RID0695 | sub-RID0827 | sub-RID0919 |
| sub-RID0020                               | sub-RID0054 | sub-RID0186   | sub-RID0175 | sub-RID0241            | sub-RID0286 | sub-RID0307 | sub-RID0338 | sub-RID0420 | sub-RID0508                 | sub-RID0588     | sub-RID0647 | sub-RID0700 | sub-RID0830 | sub-RID0927 |
| sub-RID0021                               | sub-RID0055 | sub-RID0112   | sub-RID0179 | sub-RID0250            | sub-RID0287 | sub-RID0309 | sub-RID0356 | sub-RID0424 | sub-RID0520                 | sub-RID0589     | sub-RID0648 | sub-RID0785 | sub-RID0833 |             |
| sub-RID0024                               | sub-RID0058 | sub-RID0117   | sub-RID0186 | sub-RID0252            | sub-RID0288 | sub-RID0317 | sub-RID0365 | sub-RID0440 | sub-RID0522                 | sub-RID0595     | sub-RID0649 | sub-RID0786 | sub-RID0839 |             |
| sub-RID0030                               | sub-RID0060 | sub-RID0131   | sub-RID0190 | sub-RID0259            | sub-RID0289 | sub-RID0320 | sub-RID0371 | sub-RID0442 | sub-RID0529                 | sub-RID0596     | sub-RID0650 | sub-RID0815 | sub-RID0854 |             |
| sub-RID0031                               | sub-RID0063 | sub-RID0139   | sub-RID0192 | sub-RID0267            | sub-RID0290 | sub-RID0322 | sub-RID0381 | sub-RID0452 | sub-RID0530                 | sub-RID0600     | sub-RID0652 | sub-RID0816 | sub-RID0860 |             |
| sub-RID0032                               | sub-RID0064 | sub-RID0143   | sub-RID0193 | sub-RID0272            | sub-RID0291 | sub-RID0325 | sub-RID0382 | sub-RID0454 | sub-RID0536                 | sub-RID0602     | sub-RID0656 | sub-RID0817 | sub-RID0884 |             |
| sub-RID0033                               | sub-RID0065 | sub-RID0146   | sub-RID0194 | sub-RID0274            | sub-RID0292 | sub-RID0328 | sub-RID0385 | sub-RID0459 | sub-RID0540                 | sub-RID0603     | sub-RID0658 | sub-RID0818 | sub-RID0887 |             |
| sub-RID0037                               | sub-RID0068 | sub-RID0157   | sub-RID0206 | sub-RID0278            | sub-RID0294 | sub-RID0329 | sub-RID0386 | sub-RID0472 | sub-RID0562                 | sub-RID0604     | sub-RID0677 | sub-RID0819 | sub-RID0894 |             |
| nishants@pioneer:/data/Human_Data/ cnt_1s |             |               |             |                        |             |             |             |             |                             |                 |             |             |             |             |

# Current state of data organising

```

[nishants@cnt1 project]$ ls
3t_projects  cnt_dacdatadrop  icu_streaming  imaging_process  nlp  projects_noPHI
7t_projects  cnt_dacdatadrop  ieeg_projects  imaging_projects  pioneer  rns
[nishants@cnt1 project]$ cd imaging_process/
[nishants@cnt1 imaging_process]$ ls
3T_819126  7T_818407  dicom_receiver  Flywheel_proj_DavisUnsorted  programs  tmp_transfer
[nishants@cnt1 imaging_process]$ cd programs/
[nishants@cnt1 programs]$ ls
dcm2niix_queue_prevent.txt  edit_json_queue.txt  MEG.sh  redcap.py  sort_dicoms.py
dcm2niix_queue.txt  edit_json.sh  mne-bids-pipeline-main  redcap_queue.txt  sort_dicoms_queue.txt
delete_later  heudiconv_queue.txt  prevent  redcap.sh  sort_dicoms.sh
edit_json_prevent.py  heudiconv.sh  protocols_BIDS_20220601.csv  run_dcm2niix_prevent.sh  unzip.sh
edit_json_prevent.sh  heuristic.py  protocols_BIDS.csv  run_dcm2niix.sh  validate_bids.py
edit_json.py  MEG_queue.txt  __pycache__  sandbox
[nishants@cnt1 programs]$

```

## CNT1

|                    |                           |                    |                  |                    |                                   |
|--------------------|---------------------------|--------------------|------------------|--------------------|-----------------------------------|
| 3T_metadata        | 4/11/2022 10:15 AM        | File folder        | 3T_raw           | 5/20/2022 9:45 AM  | File folder                       |
| 7T_CEST            | 10/25/2022 12:03 ...      | File folder        | 7T_raw           | 10/7/2021 9:39 AM  | File folder                       |
| 7T_metadata        | 10/7/2021 9:47 AM         | File folder        | CAPES            | 1/23/2023 11:03 AM | File folder                       |
| chop_fs            | 10/7/2021 12:39 PM        | File folder        | CHOP             | 4/14/2022 3:29 PM  | File folder                       |
| collaborative_fs   | 10/7/2021 12:26 PM        | File folder        | clinical_img_raw | 11/18/2022 4:16 PM | File folder                       |
| data_eco           | 1/18/2023 12:58 AM        | File folder        | ESI              | 5/20/2022 11:55 AM | File folder                       |
| eeg_raw            | 1/18/2023 12:49 AM        | File folder        | HUP              | 1/18/2023 11:30 AM | File folder                       |
| eip_cleanup        | 12/1/2021 3:34 PM         | File folder        | MUSC             | 5/24/2022 1:27 PM  | File folder                       |
| EMU NLP            | 5/25/2022 4:07 PM         | File folder        | RID0030          | 11/3/2022 11:52 AM | File folder                       |
| ieeg_metadata      | 1/8/2023 7:23 PM          | File folder        | RID0186          | 12/5/2022 11:33 AM | File folder                       |
| imaging_process_fs | 9/19/2022 5:52 PM         | File folder        | RID0241          | 6/8/2022 8:53 PM   | File folder                       |
| <b>imaging_raw</b> | <b>1/18/2023 12:50 AM</b> | <b>File folder</b> | RID0583          | 6/6/2022 8:55 AM   | File folder                       |
| nlp_fs             | 8/3/2022 11:26 AM         | File folder        | RID595           | 7/22/2022 4:12 PM  | File folder                       |
| pioneer_fs         | 2/11/2022 10:12 AM        | File folder        | RID864           | 10/19/2022 3:37 PM | File folder                       |
| projects_noPHI     | 9/23/2022 4:30 PM         | File folder        | RID864           | 10/19/2022 9:29 PM | Compressed (zipp... 2,930,649 KB) |
| rns_fs             | 8/1/2022 4:18 PM          | File folder        |                  |                    |                                   |
| staff_admin        | 11/22/2022 5:26 PM        | File folder        |                  |                    |                                   |

## CNT-fs

# Current state of SOP: DICOM to NIFTI

## Clinical BIDS Naming Schema

Clinical scans, those ordered as a part of standard clinical care for patients, will be saved for research subjects, per IRB protocol. Each clinical will be organized into their own BIDS session with the following naming schema:

sub-RID#### /

ses-[DateOfScan]clin[Timepoint][TypeofScan] /

{Data Type} /

sub-RID####\_ses-[DateOfScan]clin[Timepoint][TypeofScan]\_acq-"hard coded sequence name"\_{modality}.extension

sub-RID####

ses-[DateOfScan]clin[Timepoint][TypeofScan]

Data Type (ex: anat, func, ieeg, eeg, ct, pet, etc.)

sub-RID####\_ses-[DateOfScan]clin[Timepoint][TypeofScan]

sub-RID1234\_ses-20200101clin00MRI\_acq-"hard coded seq

sub-RID1234\_ses-20200101clin00MRI\_acq-"hard coded seq

Clinical Imaging Workflow

## Clinical Imaging Processing SOP

### Definitions:

**Type of Scan:** mri, meg, eeg, ct, pet

**Data type** - a functional group of different types of data. BIDS defines eight data mapping data such as field maps), anat (structural imaging such as T1, T2, P electroencephalography), beh (behavioral). Data files are contained in a direct directories.

### Export clinical raw data from PACS/SECTRA

TO DO: write down steps to export data, write down any export settings selected

1. Log on to UPHS computer remotely (kitchen computer)
2. Mount CNT-fs in File Explorer
3. Go into imaging\_raw/clinical\_imaging\_raw
4. Make an RID folder for patient (RID###)
5. Log into PennChart and go into patient's chart
6. Go into their Sectra
7. Select image of interest and hit "Export to Media"
8. Uncheck: 'Include DICOM viewer' and 'Include annotations'
9. Save images to cnt-fs/imaging\_raw/clinical\_img\_raw

### Sort Dicom processing

Next, run the sort dicoms program in cnt1 to process the raw clinical data into human-readable source Dicom data and save output into cnt-fs.

1. Open the VDI and launch MobaXterm
2. In MobaXterm, ssh into cnt1 by typing "**ssh username@cnt1**", enter your PMACS password to log in, and navigate to **project/imaging\_process/programs**
  1. In the file explorer to the left in MobaXterm, check the "Follow terminal folder" box so that the file explorer follows you as you navigate in the terminal
3. In the file explorer to the left, right click on **sort\_dicoms\_queue.txt** and click "Open with default text editor", and the MobaXterm text editor should open
4. In the text file, for each RID you want to process, type the source directory, output directory and RID name in the following format: **/mnt/cnt-fs/imaging\_process\_fs/dicom/ sub-RIDXXXX /mnt/cnt-fs/imaging\_process\_fs/dicom/ sub-RIDXXXX**
  1. Make sure to put spaces in between each input (doesn't matter how many spaces you put)
  2. Make sure to start a new line for each RID you want to process
  3. Make sure to save the text file once you are finished
5. Go back to the terminal and type "**module load python**" and hit enter, then type "**bash sort\_dicoms.sh**" and hit enter to run the program
6. Once the program is finished, navigate to the output directory (for example: **cnt-fs/imaging\_process\_fs/dicom**) and review any Dicom with conversion issues
  1. A **sub-RIDXXXX\_flagged** folder will potentially be generated along with the sub-RIDXXXX folder, containing files that were flagged to have issues during the sort Dicom process. Manually review each file, and if you would like the file to be included in the conversion pipeline, move the file to its corresponding folder in **sub-RIDXXXX**. After you are finished, move the flagged folder and any remaining contents into the **delete\_later** folder.
7. Once finished, go to the Dicom sub-RIDXXXX folder and manually rename each session folder (currently named by study date) according to the CNT clinical BIDS naming schema: **ses-[DateOfScan]clin[Timepoint][TypeofScan]**
  1. See "Clinical BIDS naming schema" in LabArchives for more information

### Run Heudiconv program

## Suggested next steps

- Sync BIDS directory in **Leif** and **BSC** clusters
- **Data already collected:** Make a sheet of data that is already organised on Leif and BSC cluster and identify missing data.
- Copy BIDS folder from Leif/BSC cluster on CNT-fs.
  - Move all source data to source folder in CNT-fs
  - Delete source data from Leif/BSC cluster
- Delete files on CNT-fs that are old/redundant
- Delete files on CNT-1 that are old/redundant
- All RC should learn how to convert files from dicom to nifti to BIDS.
  - MATLAB with Nishant
  - Python version with Alfredo and Josh
  - Important: Keep this really simple!
- Keep Alfredo, Akash, Ryan, and Nishant involved.

- Clean CNT-1 and CNT-fs [**Research coordinators (Josh lead)**]
  - Make folder BIDS + Source Data
- Sync Pioneer and BSC → [**Research people**]
  - Find missing files
  - Making ReadCap → [**Akash, Josh**]
- Harmonise BIDS converter in Python → [**Alfredo and Josh**]
  - Get 3 example dicom raw files and train → [**AT, MM, MD, BP**]
- Download missing files → [**AT, MM, MD, BP**]
- Derivative
  - Setup iEEG recon on CNT-1 → also Penn+Box
  - Recon-all output